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Verification, Validation, and Calibration of SDLC Survey Links

1. Introduction

Verification, Validation, and Calibration are important concepts in software measurement that help ensure any evaluation is accurate, consistent, and meaningful.

- Verification is about checking that the evaluation process is applied correctly.
- Validation focuses on making sure we are measuring the right thing.
- Calibration means adjusting the evaluation when we notice inconsistencies or errors.

In this assignment, I evaluated three SDLC survey links over several rounds using an evaluation matrix. The main goal was to improve the reliability and validity of my evaluation by repeating the process and refining my judgments over time.

2. Methodology

To keep the evaluation consistent, I used the same evaluation matrix for all three links across five rounds.

Evaluation Criteria (Scale: 1–5)

- Design & Colors: how visually comfortable and consistent the interface is
- Language & Clarity: how easy the instructions are to understand
- Quality: how well the content is organized and how useful it is

Process:

- I evaluated each link five times
- After each round, I recorded my scores and wrote short notes
- When I noticed inconsistencies, I adjusted my evaluation

This process helped me reduce bias and make my evaluation more accurate.

3. Evaluation Results

Link 1 (Basic)

Evaluation Criteria	Round 1:	Round 2:	Round 3:	Round 4:	Round 5:
Design & Colors	4	3	2	2	2
Language & Clarity	4	4	4	3	3
Quality	5	5	4	4	4
Key Measurement : 1: Low 3: Neutral 5: Excellent					

Justification:

At first, the design looked good, but after repeating the evaluation, I realized that the use of too many colors made the interface a bit distracting. This affected my focus while reviewing it.

The instructions were mostly clear, but some parts were slightly confusing, which is why the clarity score decreased in later rounds.

The overall quality was good because the structure followed SDLC steps, but after a deeper look, I slightly reduced the score to reflect minor issues.

Interpretation:

The scores became more stable over time, which shows better reliability. The adjustments I made also helped improve validity, since my evaluation became more accurate.

Link 2 (Normal)

Evaluation Criteria	Round 1:	Round 2:	Round 3:	Round 4:	Round 5:
Design & Colors	3	3	3	2	2
Language & Clarity	3	3	3	3	2
Quality	4	4	4	4	3
Key Measurement : 1: Low 3: Neutral 5: Excellent					

Justification:

The design felt average at first, but over time, the colors started to feel overwhelming and distracting.

The language was understandable, but I noticed that there weren't enough detailed explanations for each step, which made it harder to fully understand the purpose behind them.

The quality was acceptable, but it lacked depth compared to what I expected.

Interpretation:

The scores were fairly consistent, which indicates reasonable reliability. Small adjustments improved the validity of the evaluation.

Link 3 (Advanced)

Evaluation Criteria	Round 1:	Round 2:	Round 3:	Round 4:	Round 5:
Design & Colors	4	3	3	2	2
Language & Clarity	4	4	3	3	3
Quality	4	4	3	4	5
Key Measurement : 1: Low 3: Neutral 5: Excellent					

Justification:
This link had the same issue with colors being distracting, especially after repeated evaluations.
The instructions were generally clear, but some steps needed more explanation to make everything fully understandable.
Over time, I realized that the structure was actually strong and well-organized, which is why I increased the quality score in the final round.

Interpretation:
After calibration, my evaluation became more accurate, leading to better validity and consistent reliability.

4. Adjustment Logbook

Round	Link	Adjustment	Reason	Impact
2	All	Re-evaluated design	Colors were more distracting than expected	More accurate scoring
3	Link 1	Adjusted quality	Better understanding after re-evaluation	Improved validity
3	Link 2	Adjusted clarity	Noticed lack of explanations	More precise evaluation
3	Link 3	Adjusted quality	Recognized stronger structure	Increased accuracy
4	All	Standardized scoring	To reduce inconsistency	Improved reliability
5	All	No major changes	Scores became stable	High consistency

5. Final Analysis

At the beginning, my evaluation was more based on first impressions. However, as I repeated the process, I started to notice patterns and refine my scoring.

Calibration played an important role in improving both reliability and validity:

- My scores became more consistent across rounds
- My judgments became more accurate

Comparison:

- Link 1: Good structure but affected by visual design issues
- Link 2: Average overall with some clarity limitations
- Link 3: Strongest in terms of structure and overall quality

6. Conclusion

This assignment helped me understand how important it is to evaluate systems in a structured and repeated way.

- Verification helped me stay consistent in how I applied the evaluation
- Validation ensured that I was measuring the right aspects
- Calibration allowed me to improve my evaluation over time

By the final rounds, my evaluation became more stable, consistent, and accurate. Overall, this process showed how repeated assessment can improve the quality of measurement in software engineering.